Homework

Problem 1

We're looking at the piecewise function:

$$f(x) = \begin{cases} \frac{2x^3-5x}{x^3+1} & \text{if } x < 2 \\ \frac{x^2-x-2}{x-2} & \text{if } x > 2 \end{cases}$$

(a) Let's find the one-sided limits as x approaches 2.

- **Approaching from the left ($x \to 2^-$):** We use the top part of the function since x is less than 2. Plugging in 2 gives us:

$$\lim_{x \to 2^-} f(x) = \frac{2(2)^3 - 5(2)}{(2)^3 + 1} = \frac{16 - 10}{8 + 1} = \frac{6}{9} = \frac{2}{3}$$

- **Approaching from the right ($x \to 2^+$):** We use the bottom part. If we plug in 2, we get a divide-by-zero issue, so let's factor the top first to cancel things out:

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} \frac{(x-2)(x+1)}{x-2} = \lim_{x \to 2^+} (x+1) = 2 + 1 = 3$$

(b) Does the overall limit at $x = 2$ exist?

No, it doesn't. For a normal limit to exist, both sides have to meet at the same place. Since our left limit ($\frac{2}{3}$) and our right limit (3) are totally different, the limit does not exist.

(c) Now for the end behaviors (limits at infinity).

- **As $x \to -\infty$:** We're on the left side of the piecewise function. We just look at the highest powers of x:

$$\lim_{x \to -\infty} \frac{2x^3 - 5x}{x^3 + 1} = \lim_{x \to -\infty} \frac{x^3(2 - \frac{5}{x^2})}{x^3(1 + \frac{1}{x^3})} = \frac{2}{1} = 2$$

- **As $x \to \infty$:** We're on the right side. Using the simplified version we found in part (a):

$$\lim_{x \to \infty} \frac{x^2 - x - 2}{x - 2} = \lim_{x \to \infty} (x + 1) = \infty$$

(d) Let's find the discontinuities.

1. There's an issue at $x = -1$ in the top function because the denominator ($x^3 + 1$) becomes zero, but the numerator doesn't. This creates a vertical asymptote, which is an **infinite discontinuity**.
2. At $x = 2$, the graph jumps from a y-value of $\frac{2}{3}$ to 3. This is a classic **jump discontinuity**.

Problem 2

(a) For $f(x) = \frac{\frac{1}{(x-1)^2} - \frac{1}{4}}{x+1}$, finding the limit at $x = -1$ requires some cleanup first to deal with the complex fraction. Let's get a common denominator for the top part:

$$\text{Numerator} = \frac{4 - (x-1)^2}{4(x-1)^2} = \frac{4 - (x^2 - 2x + 1)}{4(x-1)^2} = \frac{-x^2 + 2x + 3}{4(x-1)^2}$$

Now, if we factor that quadratic on top, we get $-(x-3)(x+1)$. Putting it all back together:

$$f(x) = \frac{-(x-3)(x+1)}{4(x-1)^2(x+1)} = \frac{-(x-3)}{4(x-1)^2}$$

Now we can easily plug in -1:

$$\lim_{x \to -1} f(x) = \frac{-(-1-3)}{4(-1-1)^2} = \frac{4}{4(-2)^2} = \frac{4}{16} = \frac{1}{4}$$

(b) Is $g(x) = \frac{x^3 - 8}{x^2 - 4}$ continuous at $x = 2$?

No, because plugging in 2 gives us $\frac{0}{0}$, meaning the function is undefined there. However, we can factor both the top (difference of cubes) and bottom (difference of squares):

$$\lim_{x \to 2} \frac{(x-2)(x^2 + 2x + 4)}{(x-2)(x+2)} = \lim_{x \to 2} \frac{x^2 + 2x + 4}{x+2} = \frac{12}{4} = 3$$

Because the limit exists, we have a hole in the graph. This is a **removable discontinuity**, and we could fix it just by defining $g(2) = 3$.

(c) We need to find c to make $h(x) = \begin{cases} c^2 x + 2 & \text{if } x \leq 2 \\ cx^2 + 6 & \text{if } x > 2 \end{cases}$ continuous everywhere.

Basically, we just need the two pieces to glue together perfectly at $x = 2$. So, we set their limits equal to each other at that point:

- Left side: $c^2(2) + 2 = 2c^2 + 2$
- Right side: $c(2)^2 + 6 = 4c + 6$

Set them equal and solve the quadratic:

$$2c^2 + 2 = 4c + 6 \implies 2c^2 - 4c - 4 = 0 \implies c^2 - 2c - 2 = 0$$

Using the quadratic formula gives us our answers for c:

$$c = \frac{2 \pm \sqrt{4 - 4(1)(-2)}}{2} = \frac{2 \pm \sqrt{12}}{2} = 1 \pm \sqrt{3}$$

(d) Showing $x^4 - 4x^3 + 2 = 0$ has a solution between 0 and 1.

This is a job for the Intermediate Value Theorem. Let $P(x) = x^4 - 4x^3 + 2$. Since it's a polynomial, it's nice and continuous.

Let's check the endpoints: $P(0) = 2$ (which is positive) and $P(1) = -1$ (which is negative).

Because the graph has to go from a positive y-value down to a negative y-value without lifting the pen, it absolutely has to cross the x-axis (where $y = 0$) somewhere between $x = 0$ and $x = 1$.

Problem 3

(a) $\lim_{x \to 0} \frac{1 - \cos(5x)}{3x^2}$

To fix this, we can multiply the top and bottom by the conjugate, $1 + \cos(5x)$:

$$= \lim_{x \to 0} \frac{1 - \cos^2(5x)}{3x^2(1 + \cos(5x))} = \lim_{x \to 0} \frac{\sin^2(5x)}{3x^2(1 + \cos(5x))}$$

Now we can group things to use the special limit $\lim_{x \to 0} \frac{\sin(ax)}{ax} = 1$:

$$= \lim_{x \to 0} \left(\frac{25}{3} \right) \cdot \left(\frac{\sin(5x)}{5x} \right)^2 \cdot \frac{1}{1 + \cos(5x)} = \frac{25}{3} \cdot (1)^2 \cdot \frac{1}{2} = \frac{25}{6}$$

(b) $\lim_{x \to -3} \frac{x+3}{\sqrt{x^2+7}-4}$

Let's get rid of the root in the denominator by rationalizing (multiplying top and bottom by $\sqrt{x^2+7}+4$):

$$= \lim_{x \to -3} \frac{(x+3)(\sqrt{x^2+7}+4)}{x^2+7-16} = \lim_{x \to -3} \frac{(x+3)(\sqrt{x^2+7}+4)}{x^2-9}$$

Factor the bottom to $(x-3)(x+3)$ and cancel out the problem term:

$$= \lim_{x \to -3} \frac{\sqrt{x^2+7}+4}{x-3} = \frac{\sqrt{16}+4}{-3-3} = \frac{8}{-6} = -\frac{4}{3}$$

(c) $\lim_{x \to -\infty} \frac{\sqrt{25x^2+3x}}{4x-1}$

Here we factor out the highest power, but we have to be super careful! Since $x \to -\infty$, when we pull x^2 out of the square root, it becomes $|x|$, which is $-x$.

$$\lim_{x \to -\infty} \frac{-x\sqrt{25 + \frac{3}{x}}}{x(4 - \frac{1}{x})} = \lim_{x \to -\infty} \frac{-\sqrt{25 + \frac{3}{x}}}{4 - \frac{1}{x}} = \frac{-\sqrt{25}}{4} = -\frac{5}{4}$$

(d) $\lim_{x \to 0} x^4 \sin\left(\frac{\pi}{x}\right)$

This one is perfect for the Squeeze Theorem. We know sine always bounces between -1 and 1, no matter how crazy the inside gets:

$$-1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1$$

Multiply everything by x^4 (which is safe because x^4 is always positive):

$$-x^4 \leq x^4 \sin\left(\frac{\pi}{x}\right) \leq x^4$$

As x goes to 0, both $-x^4$ and x^4 go to 0. Since our function is squeezed between two things going to 0, its limit must also be 0.

Problem 4

We're working with $f(x) = x^3 - 6x^2 + 9x$ and $g(x) = \sqrt{3x+4}$.

(a) Tangent line of f at $x = 1$.

First, find the coordinate: $f(1) = 1 - 6 + 9 = 4$, so our point is $(1, 4)$.

Next, find the slope by taking the derivative: $f(x) = 3x^2 - 12x + 9$.

At $x = 1$, the slope is $f(1) = 3(1)^2 - 12(1) + 9 = 0$.

A line with a slope of 0 is just a horizontal line. Using point-slope form, it simplifies right down to $y = 4$.

(b) Average rate of change vs. instantaneous at $x = 1$.

Average rate on $[0, 2]$ is just the slope of the secant line: $\frac{f(2)-f(0)}{2-0}$.

$f(2) = 2$, and $f(0) = 0$. So, the average rate is $\frac{2-0}{2} = 1$.

We already know from part (a) that the instantaneous rate at $x = 1$ is 0.

They are different, which makes sense because the curve isn't a straight line.

(c) Finding $g(x)$ using the limit definition.

Here's the setup with the difference quotient:

$$g(x) = \lim_{h \to 0} \frac{\sqrt{3(x+h)+4} - \sqrt{3x+4}}{h}$$

We rationalize the numerator to get rid of the roots on top:

$$= \lim_{h \to 0} \frac{(3x+3h+4) - (3x+4)}{h(\sqrt{3x+3h+4} + \sqrt{3x+4})}$$

The top simplifies beautifully to just $3h$, which lets us cancel the h on the bottom:

$$= \lim_{h \to 0} \frac{3h}{h(\sqrt{3x+3h+4} + \sqrt{3x+4})} = \lim_{h \to 0} \frac{3}{\sqrt{3x+3h+4} + \sqrt{3x+4}}$$

Now, just plug in $h = 0$ to get the final derivative: $\frac{3}{2\sqrt{3x+4}}$.

(d) Normal line of g at $x = 4$.

The normal line is just perpendicular to the tangent line.

The point is $g(4) = \sqrt{12+4} = 4$, giving us $(4, 4)$.

The tangent slope is $g(4) = \frac{3}{2\sqrt{16}} = \frac{3}{8}$.

To get the perpendicular normal slope, we flip it and reverse the sign: $= -\frac{8}{3}$.

Put it into point-slope form: $y - 4 = -\frac{8}{3}(x - 4)$. Distributing and solving for y gives us $y = -\frac{8}{3}x + \frac{44}{3}$.

Problem 5

For the curve $3x^2 - 2xy + 2y^2 = 14$:

(a) Let's find $\frac{y}{x}$ implicitly. Don't forget the product rule for the $-2xy$ term!

Taking the derivative with respect to x:

$$6x - (2y + 2xy) + 4yy = 0$$

Distribute the negative and group all the terms with y on one side:

$$6x - 2y - 2xy + 4yy = 0 \implies y(4y - 2x) = 2y - 6x$$

Divide to isolate y:

$$y = \frac{2y - 6x}{4y - 2x} = \frac{y - 3x}{2y - x}$$

(b) Horizontal tangents.

These happen when the slope is zero, meaning the top of our derivative fraction is zero: $y - 3x = 0$, so $y = 3x$.

Plug $y = 3x$ back into the original curve equation:

$$3x^2 - 2x(3x) + 2(3x)^2 = 14 \implies 3x^2 - 6x^2 + 18x^2 = 14$$

$$15x^2 = 14 \implies x = \pm\sqrt{\frac{14}{15}}$$

Since y is 3 times x, the points are $\left(\sqrt{\frac{14}{15}}, 3\sqrt{\frac{14}{15}}\right)$ and $\left(-\sqrt{\frac{14}{15}}, -3\sqrt{\frac{14}{15}}\right)$.

(c) Vertical tangents.

These happen when the derivative is undefined, meaning the bottom of the fraction is zero: $2y - x = 0$, so $x = 2y$.

Plug $x = 2y$ into the original equation:

$$3(2y)^2 - 2(2y)y + 2y^2 = 14 \implies 12y^2 - 4y^2 + 2y^2 = 14$$

$$10y^2 = 14 \implies y^2 = \frac{7}{5} \implies y = \pm\sqrt{\frac{7}{5}}$$

Since x is 2 times y, the points are $\left(2\sqrt{\frac{7}{5}}, \sqrt{\frac{7}{5}}\right)$ and $\left(-2\sqrt{\frac{7}{5}}, -\sqrt{\frac{7}{5}}\right)$.

(d) Finding the second derivative, $\frac{^2y}{x^2}$.

This requires applying the quotient rule to $y = \frac{y-3x}{2y-x}$:

$$y'' = \frac{(y-3)(2y-x) - (y-3x)(2y-1)}{(2y-x)^2}$$

If you expand the whole top out carefully, a bunch of stuff cancels:

$$\text{Numerator} = 2yy - xy - 6y + 3x - (2yy - y - 6xy + 3x) = 5xy - 5y = 5(xy - y)$$

Now, we can plug our first derivative formula back into $(xy - y)$:

$$x\left(\frac{y-3x}{2y-x}\right) - y = \frac{xy - 3x^2 - 2y^2 + xy}{2y-x} = \frac{-(3x^2 - 2xy + 2y^2)}{2y-x}$$

Notice that the top of that fraction is exactly the negative version of our original curve equation! So we can replace $(3x^2 - 2xy + 2y^2)$ with 14:

$$xy - y = \frac{-14}{2y-x}$$

Substitute this all back into our y'' formula for the final answer:

$$y'' = \frac{5\left(\frac{-14}{2y-x}\right)}{(2y-x)^2} = \frac{-70}{(2y-x)^3}$$